

Computer Science & IT

Discrete Mathematics



Set Theory & Algebra

Lecture No. 13



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Recap of Previous Lecture



Topic

Hasse diagram / POSET diagram



Topics to be Covered



Topic

Bounded lattice



Topic

Complements of an element in a lattice

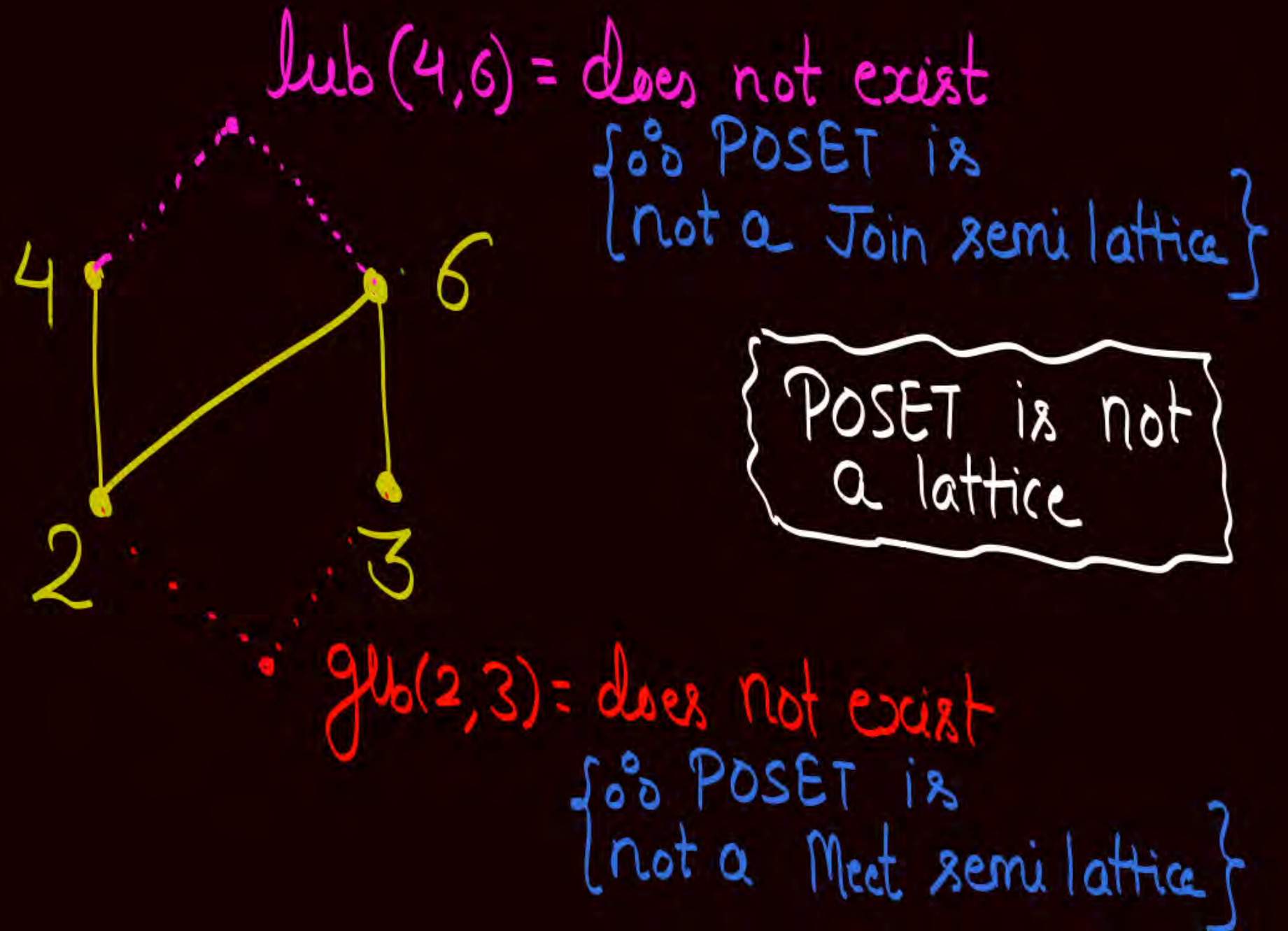


Topic

Complemented lattice



Q (iv) Construct the Hasse diagram for POSET,
 $(\{2, 3, 4, 6\}, \div)$



Q (V) Construct the Hasse diagram for POSET,
 $(\{2, 3, 6, 12\}, \div)$

* lub exist for every
Pair of elements

∴ POSET is a
Join semi lattice



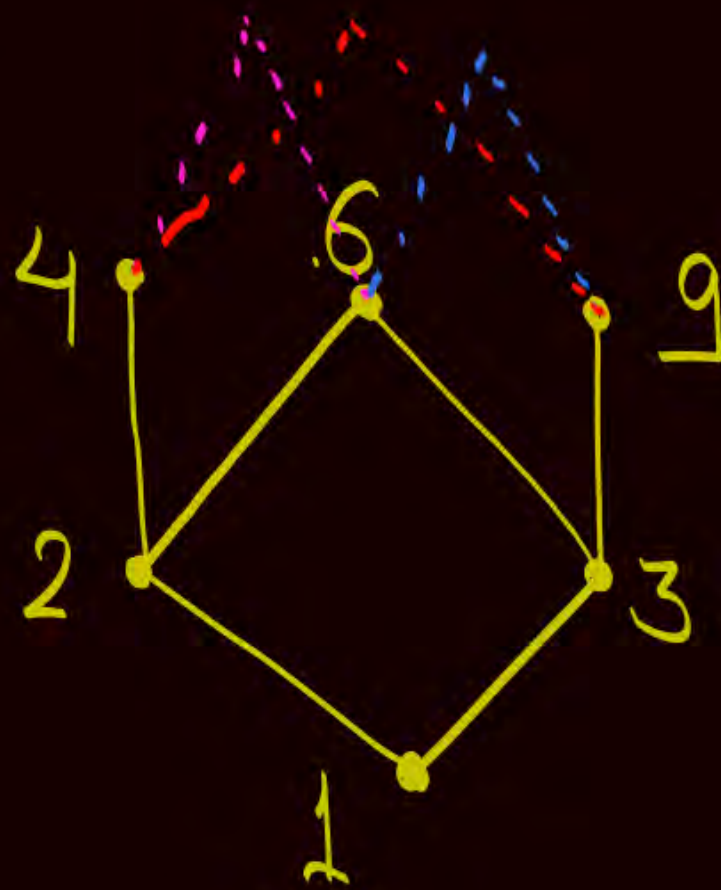
$glb(2, 3) = \text{does not exist}$

∴ POSET is not a
Meet semi lattice

∴ Not a lattice

Q (vi) Construct the Hasse diagram for POSET,
 $(\{1, 2, 3, 4, 6, 9\}, \div)$

glb exist for
Every pair of elements
∴ POSET is a
Meet semi lattice



$\text{lub}(4, 6) = \text{Does not exist}$

∴ POSET is not a
Join semi lattice

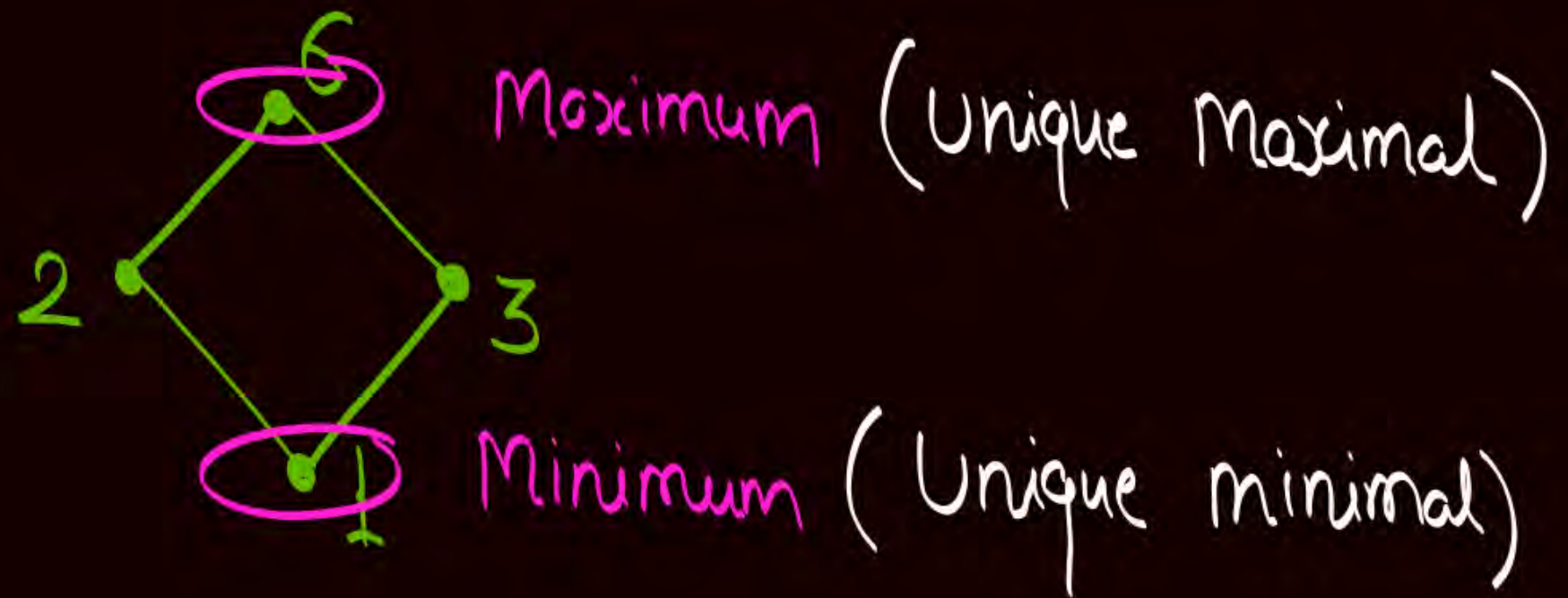
(1) $(\{-1, 0, 2.5, 4, 6\}, \leq)$



it is the maximum element
{ Unique maximal element }

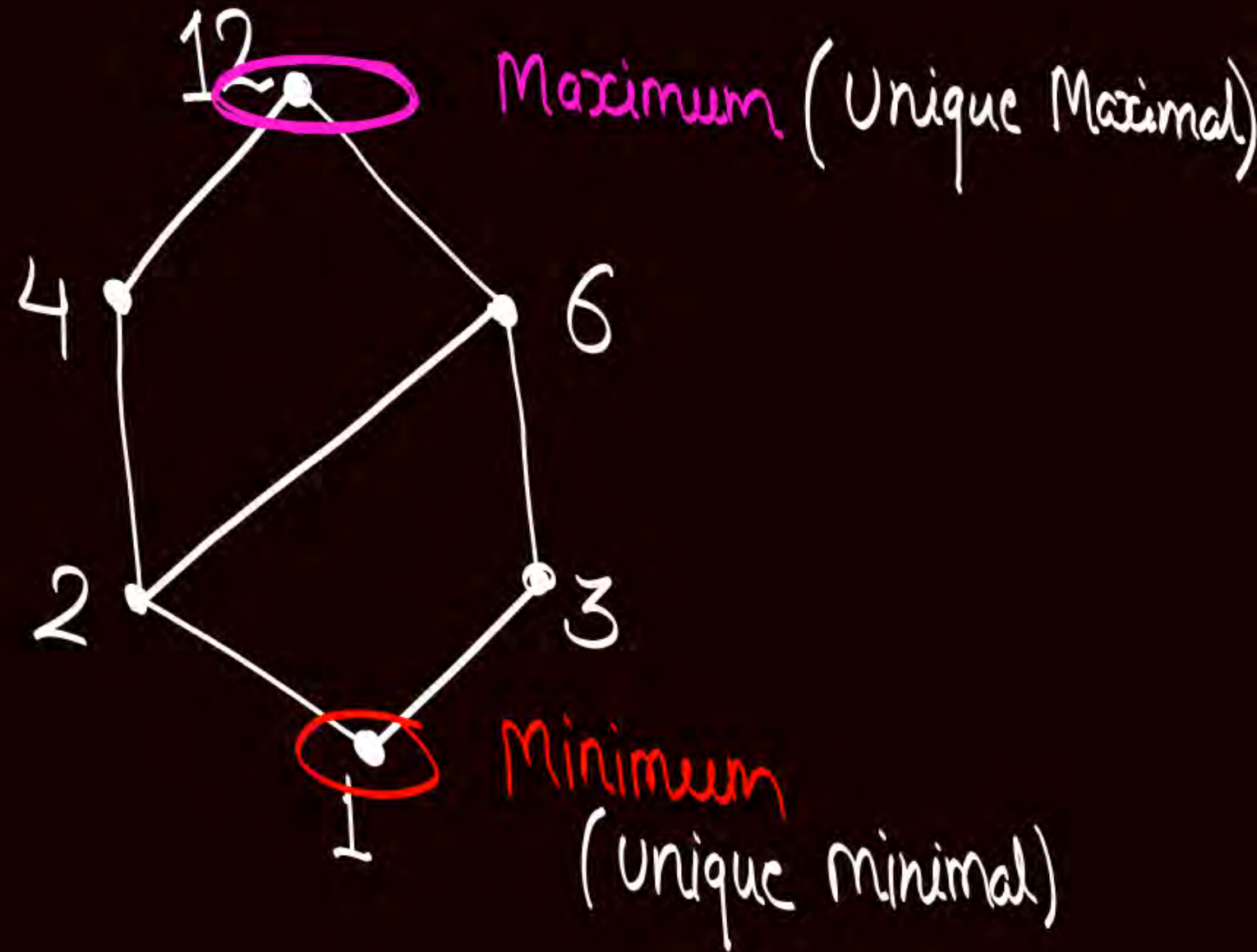
Minimum element
{ Unique minimal element }

Q (ii) Construct the Hasse diagram for POSET,
 $(D_6, \div)$

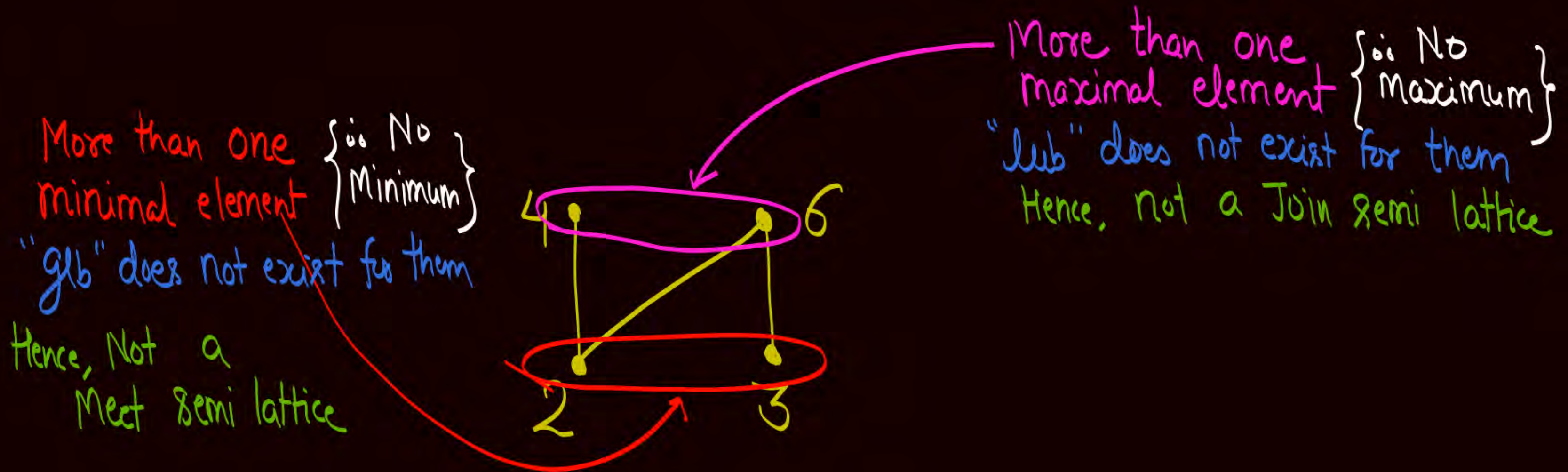


Q (iii) Construct the Hasse diagram for POSET,
 $(D_{12}, \div)$

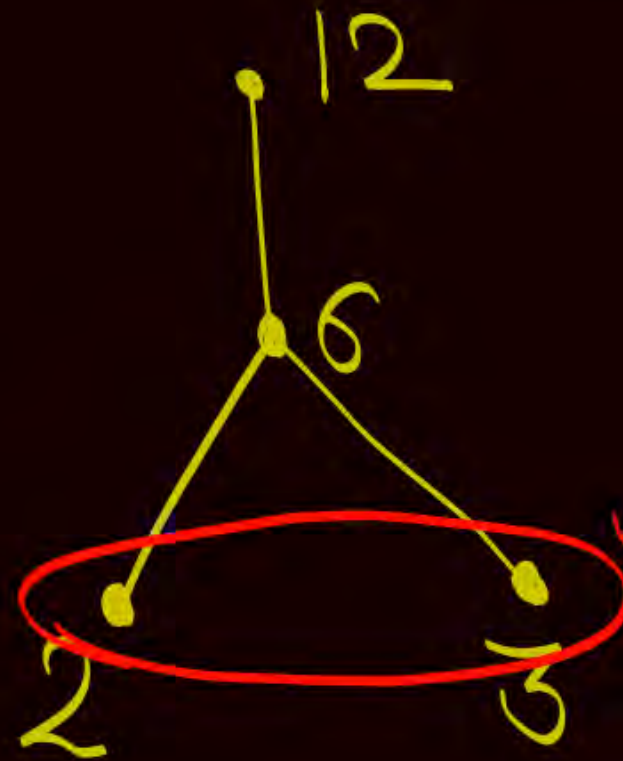
$\downarrow$
 $D_{12} = \{1, 2, 3, 4, 6, 12\}$



Q (iv) Construct the Hasse diagram for POSET,
 $(\{2, 3, 4, 6\}, \div)$

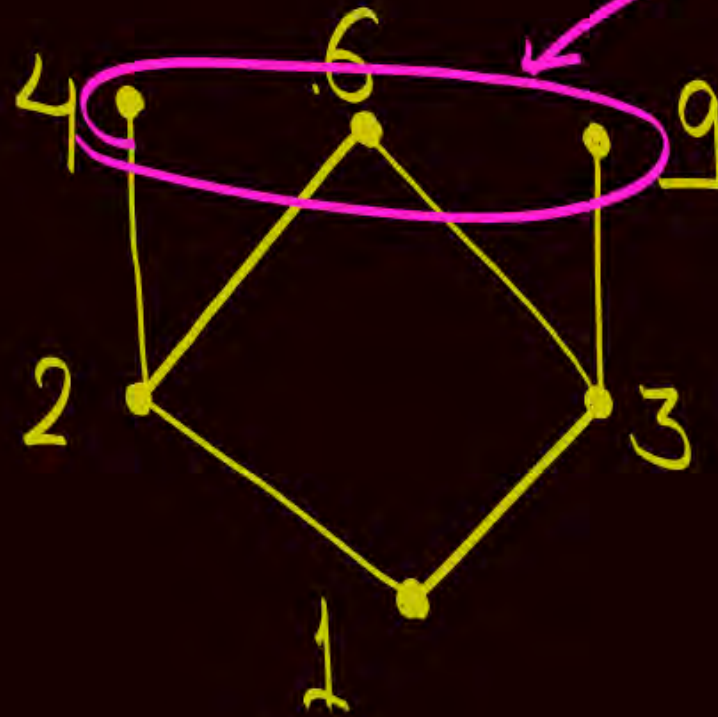


Q (V) Construct the Hasse diagram for POSET,
 $(\{2, 3, 6, 12\}, \div)$



More than one minimal
∴ Not a Meet Semi lattice

Q (vi) Construct the Hasse diagram for POSET,
 $(\{1, 2, 3, 4, 6, 9\}, \div)$

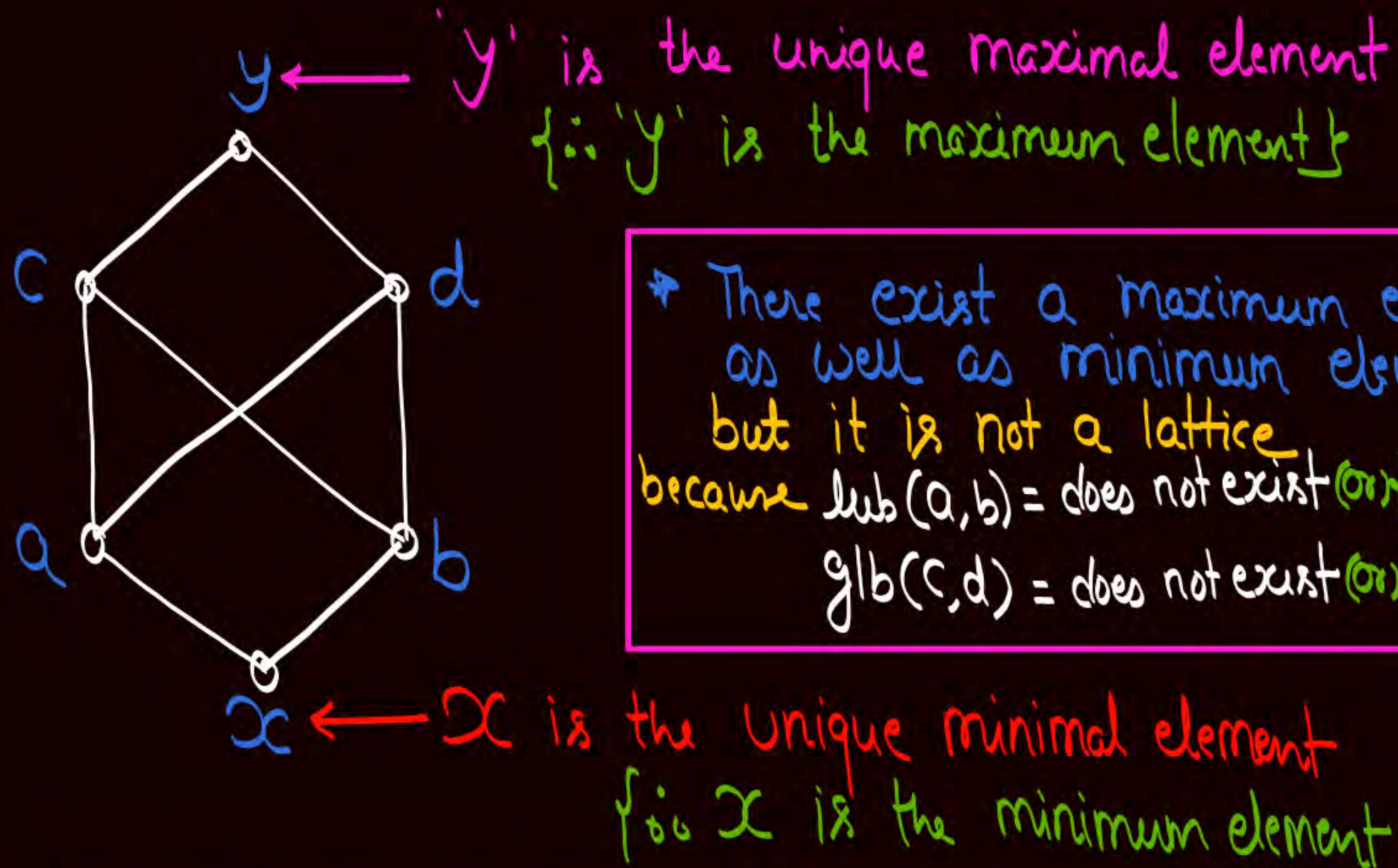


More than one maximal
∴ Not a Join Semi lattice

Note :-

- ① In a POSET if there exist two or more maximal elements then it can not be a Join semi lattice $\{ \therefore \text{Not a lattice} \}$
- ② In a POSET if there exist two or more minimal elements then it can not be a Meet semi lattice $\{ \therefore \text{Not a lattice} \}$

eg:- Consider the following POSET diagram



★ There exist a maximum element in the POSET, as well as minimum element in the POSET.

but it is not a lattice

because $\text{lub}(a, b) = \text{does not exist}$ (or) $\{\text{lub}(a, b) = \{c, d\}\}$

$\text{glb}(c, d) = \text{does not exist}$ (or) $\{\text{glb}(c, d) = \{a, b\}\}$

In a lattice lub and glb must exist and they must be unique

Note:- There may be some POSET in which there exist maximum element as well as minimum element, but still they are not lattice

Note: →

It is not necessary for a lattice to have maximum and/or minimum element, because the set may be an infinite set

$(\mathbb{N}, \leq)$

is a lattice,

Maximum element = Does not exist

Minimum element = 1

eg. $(\mathbb{Z}, \leq)$ is a POSET

Hasse diagram



* lub exist for every pair of elements as well as glb exist for every pair of elements

Hence it is a lattice,

But, maximum element does not exist

And Minimum element also does not exist.



Topic : Bounded Lattice

Let $[L, \vee, \wedge]$ be a lattice

① If there exists an element $I \in L$

$$\text{s.t. } a \vee I = I \quad \forall a \in L$$

then 'I' is called Universal upper bound

(or) Upper bound of lattice L

"I" is actually the maximum element of lattice.

② If there exist an element $O \in L$.

$$\text{Such that, } a \wedge O = O, \quad \forall a \in L$$

then, 'O' is called Universal lower bound

(or) lower bound of lattice L

'O' is actually the minimum element

* A lattice $[L, \vee, \wedge]$ is called a bounded lattice if and only if both 'I' and 'O' exist in the lattice

(or)
* A lattice is called a bounded lattice if and only if both maximum & minimum element exist.

Note:- ① Every finite lattice is a bounded lattice

② If lattice is an infinite lattice then it may or may not be a bounded lattice

eg: $(\mathbb{Z}, \leq)$ and $(\mathbb{N}, \leq)$ are example of unbounded lattice
Where both $\mathbb{Z}$ & $\mathbb{N}$ are infinite sets

Bounded lattice $\Rightarrow$ i.e. infinite set

eg: let $A = \{x \mid x \in \mathbb{R} \text{ and } 0 \leq x \leq 1\}$
 $\hookrightarrow A$ is an infinite set.

We know $(A, \leq)$ is a lattice where
'A' is an infinite set

but $0 = \text{Minimum element}$

& $1 = \text{Maximum element}$

$\rightarrow$ Both minimum & maximum element exist
so it is a bounded lattice.



Topic : Complement of an element in a lattice

* Complements can be defined for an element of a lattice L only if ' L ' is a bounded lattice

→ i.e. If lattice L is not a bounded lattice then complement does not exist for any element of lattice ' L '



Topic : Complement of an element in a lattice

→ Let $[L, \vee, \wedge]$ be a bounded lattice

Where, I = Upper bound of lattice L { i.e. Maximum element }
and O = lower bound of lattice L { i.e. Minimum element }

→ for an element $a \in L$

if there exist some element $b \in L$

such that (i) $a \vee b = I$

and (ii) $a \wedge b = O$,

then 'a' and 'b' are Complement of each other

Note :

① In a bounded lattice,
'I' and 'O' are always Complement of each other
and they do not have any other Complement.

i.e., The only Complement of Maximum element (I) is minimum element (O)
and The only Complement of minimum element (O) is maximum element (I)

② In a bounded lattice,
Complement need not exist for every element
of the lattice, and if exists it need not be unique
(i.e., some elements may have more than one Complement)

→ In a bounded lattice Number of Complements of an element
of the lattice may be '0' or '1' or more



Topic : Complemented lattice

- A bounded lattice is called Complemented lattice if and only if at least one complement exist for every element of the lattice
- In a Complemented lattice every element has '1' or 'more' complements.

H.W. :- Find the Complement of Every element (if exist)
of the following lattices.

(i) $(\{-1, 0, 2.5, 4, 6\}, \leq)$

(ii) $(D_8, \div)$

(iii) $(D_{12}, \div)$



2 mins Summary



Topic

Bounded lattice

Topic

Complements of an element in a lattice

Topic

Complemented lattice

THANK - YOU